High-Dimensional Feature Space

$\mathbf{X} \in \mathbb{R}^{N \times D}$

f_1 f_2 f_3 ... f_D $\mathbf{Y}$

Feature Matrix & Labels

Mutual Information Estimation

$I(f_j; \mathbf{Y}) = H(\mathbf{Y}) - H(\mathbf{Y} \mid f_j)$



MI-Derived Prior Vector

$\widetilde{\mathbf{m}} = \frac{\mathbf{I} - I_{\min}}{I_{\max} - I_{\min}} \in [0, 1]^D$

Knowledge-Informed Swarm Seeding

Hybrid Dual-Subpopulation Initialization

MI-Guided Seeding

$\mathbf{p}_i = \text{clip}(\widetilde{\mathbf{m}} + \boldsymbol{\varepsilon}_i,\ 0,\ 1)$

$\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0},\ \sigma^2 \mathbf{I})$

$p_{i,j} \propto \widetilde{m}_j$

Uniform Random Seeding

$\mathbf{p}_i \sim \mathcal{U}(0,\ 1)^D$

$p_{i,j} \sim \mathcal{U}(0,\ 1)$

Stochastic Exploration

Initial PSO Swarm

$\mathcal{P} = \{\mathbf{p}_1,\ \ldots,\ \mathbf{p}_N\}$

Subpopulation 1: MI-Guided

Subpopulation 2: Random

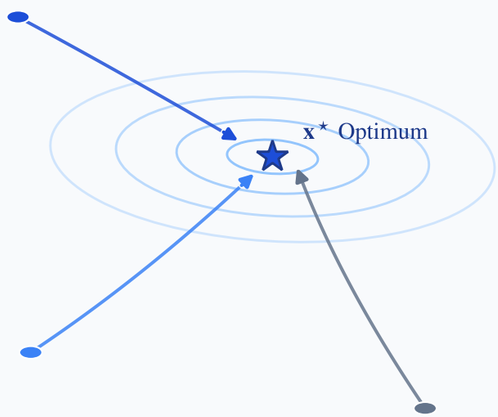
Re-inject

Adaptive MI Re-injection

$\mathbf{p}_{\text{inferior}} \leftarrow \text{clip}(\widetilde{\mathbf{m}} + \boldsymbol{\varepsilon}_i,\ 0,\ 1)$

Search Space & Recovery

Evolutionary Convergence & Resilience



Stagnation Detected

Trigger